Governed Feature Selection for Credit-Risk Models: Shapley Attribution with Pre-Registered Parsimony under the AEGIS Decision-Gate Framework

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Abstract

Machine-learning credit-risk models are increasingly accurate but remain hard to *govern*: the questions of *which* features are genuinely useful, *how parsimonious* a defensible model may be, and *whether* a selection is adequately powered and fair are usually answered by analyst judgement rather than a reproducible, auditable procedure. We present two coupled contributions. **SHARP** (Shapley Ranking with Parsimony) is a pre-registered feature-selection methodology that separates order-invariant *attribution* (sampled Shapley importance against a shadow-feature null) from a distinct, fixed *selection* rule (the smallest feature set that is both DeLong-indistinguishable from the full model and within a 5% relative-Gini tolerance), followed by always-on isotonic calibration and a regulatory PD floor. **AEGIS** is a hash-chained decision-gate framework (G0–GF) that wraps SHARP with materiality-tiered governance, binding every statistical bar to a cited source and separating *governance* decisions from *analytics* parameters, which are never re-opened. Across a corpus of eleven public and reference credit datasets run under one fixed configuration, we show that the same rule yields genuine parsimony at small sample sizes yet near-full feature sets at large ones (a large-N sensitivity effect we characterise), that excluding a protected attribute is necessary but not sufficient to remove disparate impact, and that signal strength—not sample size—sets the achievable discrimination ceiling. We add the Riley *et al.* minimum-sample-size calculation alongside events-per-variable as a reported sufficiency diagnostic and give a retrospective across the corpus.

Index Terms

Credit decisions affect access to capital for individuals and firms, and supervisors now require model-risk controls (e.g., the U.S. interagency guidance superseding SR 11-7, and the EU AI Act's treatment of creditworthiness assessment as high-risk). This work shows how feature selection—an early, consequential step usually left to discretion—can be made *reproducible and auditable* without inventing economic assumptions: thresholds and pricing remain institution inputs, and governance gates decide process, not analytics. The framework makes a model's parsimony, calibration, fairness, and data-sufficiency claims traceable to recorded runs and cited bars, which should lower the cost of independent validation and reduce the risk of overstated or unfair models reaching deployment.

Index Terms

Credit risk, feature selection, Shapley values, model governance, calibration, fairness, disparate impact, sample size, explainable AI.

I. Introduction

MACHINE-learning models now dominate credit-risk scoring, but their *governance* lags their accuracy. [Stub: motivate the three governance gaps—useful features, defensible parsimony, adequacy/fairness—and the discretion problem.]

Contributions.

1) **SHARP**, a pre-registered methodology separating Shapley *attribution* from a fixed *parsimony* selection rule (§III).
2) **AEGIS**, a hash-chained, materiality-tiered decision-gate framework binding every bar to a cited source and separating governance from analytics (§IV).
3) An empirical study over eleven datasets under one fixed configuration, including the large-N parsimony effect, the "necessary-but-not-sufficient" fairness finding, and a Riley-*et al.* sample-size retrospective (§VI).

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II. Related Work

[Stub: feature selection and Shapley/SHAP attribution; parsimony and the 1-SE heritage; model-risk governance (SR 26-2) and the EU AI Act; fairness and disparate-impact doctrine; sample size—EPV [1] vs Riley [2], [3].]

III. The SHARP Methodology

SHARP (Shapley Ranking with Parsimony) deliberately separates two questions that are routinely conflated in applied feature selection:

- **Attribution** — "How much predictive value is attributable to this feature, *order-invariantly*?" Answered by a Shapley importance measure against an empirical null.
- **Selection** — "What is the *smallest* feature set whose performance is statistically indistinguishable from the full set?" Answered by a separate, pre-registered stopping rule.

The separation is principled, not stylistic. Shapley values are a *fair-attribution* solution concept (efficiency, symmetry, dummy, additivity); they do not identify an optimal subset. A redundant-but-informative feature still receives a positive share, and a feature whose value concentrates in rare coalitions can average below any bar despite being needed. Attribution therefore *ranks*; a separately justified rule *selects*. Greedy, category-sequential screening—the common alternative—produces order-dependent, artefactual retention; an order-invariant importance estimate plus an explicit selection rule avoids it.

Inference target. We estimate population feature importance *with respect to a specified learner and value function*: how much cross-validated predictiveness is recoverable from a feature set, for a given learner class, on a given population—*not* how much a feature "matters to credit risk" in the abstract. Formally, with value function $v(S)$ equal to the cross-validated predictiveness of the columns in S, the importance of feature j is the Shapley value

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{j\}) - v(S)\big], \tag{1}$$

which is the Shapley Population Variable Importance Measure (SPVIM) of Williamson and Feng [4]. This is *not* SAGE [5], which marginalises absent features out of a single fitted model; here each coalition S is *refit*.

The pipeline has five stages (0–4). We summarise each; the governance wrapper that signs off on their outputs is AEGIS (§IV).

A. Stage 0: Eligibility and data governance

Stage 0 runs *before any statistics* and drops no column; it attaches a per-column classification that travels with every downstream result. Each column is tagged `permissible`, `prohibited` (e.g. sex, age, marital status), `proxy_risk` (e.g. geography, education, household composition), or `unclear`. On these public research benchmarks, protected and proxy attributes are *retained with mandatory disclosure* so results remain comparable with the large published literature on the same datasets; no model here is intended for live use. Stage 0 also records point-in-time availability (flagging `near_target` and `external_score` columns that make a model a behavioural or score-recovery exercise rather than an application scorecard), audits sentinel codes and shared missingness for leakage, notes the accept-only sampling that truncates feature distributions [6], and declares *mandatory* columns and a missingness policy that are enforced only at scoring (§III-F).

B. Stage 1: Univariate screening

Information Value (IV) is the single a-priori screen, at the conventional threshold $\mathrm{IV} \geq 0.02$ [7]; Cramér's V, χ^2 and p-values are reported descriptively and decide nothing. Binning is monotonic Weight-of-Evidence (WOE) with a minimum bin size of 5% of rows,

$$\mathrm{WOE}_i = \ln \frac{g_i/G}{b_i/B}, \qquad \mathrm{IV} = \sum_i \Big(\tfrac{g_i}{G} - \tfrac{b_i}{B}\Big)\mathrm{WOE}_i, \tag{2}$$

where g_i, b_i are goods/bads in bin i. Monotonicity is enforced only where it is supported: a genuinely U-shaped feature whose IV recovers $\geq 20\%$ when the constraint is freed is tagged `shape_nonmonotonic` and routed to a single-column model check rather than being judged on its monotonic IV. A reliability gate (χ^2 minimum expected cell ≥ 5) and a bin-count sensitivity check (WOE vs 5/10/20 quantile bins) escalate borderline columns to the same model check, whose verdict is a Gini confidence interval excluding zero.

C. Stage 2: Order-invariant attribution

The default value function is repeated cross-validated AUC and the default learner is a penalised logistic model on monotonic-WOE inputs (`woe_logistic`)—the credit-scoring standard form: deterministic and hyperparameter-insensitive, which removes refit noise from the Shapley estimate. XGBoost is run as a robustness learner; material disagreement in the two rankings (reported via Kendall's τ) is itself a finding about interaction dependence. Class reweighting is *off* by default: on GMSC it inflated the mean predicted probability to 0.286 against a true base rate of 0.067, and the resulting miscalibration is avoided rather than repaired.

Two devices make the attribution both tractable and honest. First, correlated columns are not pruned (which would reintroduce a greedy univariate criterion); instead, hierarchical clustering on $1 - |\rho_{\text{Spearman}}|$ cut at $|\rho| = 0.7$ makes *each cluster a single Shapley player*, preserving the efficiency property over the full candidate set. Second—the central methodological fix—significance testing of Shapley values is replaced by a **shadow-feature null** [8]: at least ten shadow players (row-permuted copies of real columns) are added, the identical Shapley computation is run over the augmented set, and a real player is `importance_confirmed` only if its value exceeds the 95th percentile of the shadow distribution *and* its bootstrap confirmation frequency is ≥ 0.80. The bar is a quantile of an empirical null, so no multiple-testing correction is needed and fold-dependence does not bias it. Benjamini–Hochberg p-values and Cohen's d are retained only as reported diagnostics—usefully self-refuting, since on German Credit a permuted-noise shadow column received $p = 0.0000$ with a *negative* Shapley value, which a significance rule would have confirmed and the shadow bar correctly rejected. Enumeration is exact when the player count (including shadows) is ≤ 15 and sampled otherwise, with the Monte Carlo standard error reported per player.

D. Stage 3: Pre-registered parsimony

The eligible set is the confirmed players, ranked by ϕ_j. For $k = 1, \ldots, K$ the model on the top-k players is scored on *out-of-fold* predictions (so every k is evaluated on the same rows and DeLong's paired test is valid [9]), and the full parsimony curve is a required output. The stopping rule is fixed in advance: choose the smallest k that is both statistically indistinguishable from the full-candidate model by DeLong's test at $\alpha = 0.05$ *and* within a relative-Gini tolerance,

$$k^{\star} = \min\Big\{k : \; p_{\text{DeLong}}(k) \geq 0.05 \;\wedge\; \tfrac{\text{Gini}_{\text{full}} - \text{Gini}_k}{\text{Gini}_{\text{full}}} \leq 0.05\Big\}. \tag{3}$$

If no k qualifies, that is reported as the result ("no parsimonious subset meets the tolerance")—it is never replaced by the best available k. Because the two conditions are combined by *and*, the rule behaves very differently across sample sizes (§VI): at small N both clear together and yield genuine parsimony, whereas at large N DeLong detects the loss from dropping even a weak feature and the full set is returned.

E. Stage 4: Calibration, reason codes, and sufficiency diagnostics

The selected model is calibrated by cross-validated isotonic regression (always on), and calibrated probabilities are used whenever a PD is quoted to a person. Because the learner is WOE-logistic, the score is an exact additive decomposition $\text{logit} = \beta_0 + \sum_j \beta_j \, \text{WOE}_j(x)$, giving faithful per-feature *reason codes* with no post-hoc approximation. A regulatory PD floor of 0.03% (Basel CRE32.3) and a 99.9% cap are applied at scoring, as are the Stage-0 mandatory-field gate and missingness policy. Finally, data sufficiency is reported—never used to auto-reject—through two diagnostics side by side: events-per-variable [1], [10] and the Riley *et al.* minimum sample size [2], [3], the latter derived from the anticipated C-statistic and prevalence via the Cox–Snell R^2 approximation of [11]. A shortfall is a disclosed caveat the governance layer weighs, not a gate failure—consistent with SHARP's principle that selection parameters are fixed and only governance decides disposition.

IV. The AEGIS Governance Framework

SHARP produces a defensible *model*; it does not, by itself, produce a defensible *decision to use that model*. A supervisor or internal audit function asks questions the analytics cannot answer: who decided the data were adequate, and could they have decided otherwise? Who chose this feature set, on what recorded rationale? Who is accountable if the model is wrong, and were they independent of the builder? Where is the evidence that each regulatory expectation was *met*, not merely claimed? AEGIS (Approval and Evidence-based Governance for

TABLE I
AEGIS DECISION GATES AND THE SHARP STAGE EACH GOVERNS.

Gate	SHARP stage	Core question decided
G0	Stage 0	Is the dataset fit to build on? (coverage, label integrity, scope)
G1	Stage 1	Which features survive screening; is N adequate (EPV + Riley, reported)?
G2	Stage 2	Is the attribution stable enough to select on?
G3	Stage 3	Which feature set is adopted—parsimonious vs full, and the trade-off?
G4	Stage 4	Calibrated, discriminating, and fair enough to approve? (per-grade backtest)
G5	Scoring	Deployment controls + cutoff policy + acceptance demonstration
GF	whole model	GO / GO-WITH-CONDITIONS / NO-GO (committee)

Internal-ratings Systems) answers these by turning each SHARP stage into a *governed decision gate*: the machine emits a recommendation and a full evidence pack, a human with the right role and sufficient independence chooses from a closed set of genuinely viable options, and the choice is written to a tamper-evident log. This mirrors modern model-risk expectations—the U.S. interagency guidance SR 26-2 (superseding SR 11-7), the EU AI Act's treatment of creditworthiness assessment as high-risk, and Basel CRE36—without importing their numeric bars by assumption: every threshold used is tied to a cited source.

A. Design principles

Eight principles govern the framework; three are load-bearing. **(P1) Machine recommends, human decides, record proves**: every gate is recommendation + evidence pack + closed options + accountable sign-off + log entry. **(P2) Segregation of duties**: the Develop, Validate, and Approve functions are never held by one person for one model. **(P8) Gates decide governance, not analytics**: a gate's option set may *never* re-open a pre-registered SHARP parameter (estimator mode, selection tolerances, shadow-bar quantile, confirmation frequency, `random_state`, CV folds). Those are fixed in the methodology, identical across every dataset, and chosen once; gates decide scope, eligibility, fairness treatment, cutoffs, and Go/No-Go. P8 is what keeps the method reportable: letting a per-run gate switch the estimator would be the dataset-dependent, path-changing behaviour a research methodology must avoid. The single failure mode AEGIS is explicitly designed against is *rubber-stamping*—hence the anti-strawman rule below and the independence requirements in §IV-E.

B. The decision gates G0–GF

SHARP's stages become gates G0–G5, with a final committee gate GF (Table I). G0–G5 are *stage gates* (a single accountable role signs each); GF is the *committee gate*, ruled by the Model Approval Committee. Crucially, GF is last and is decided only after G4's per-grade calibration backtest and G5's cutoff policy and acceptance demonstration are complete and visible—one does not approve deployment and then check whether the PDs are honest and the decisions sensible.

A model is assigned a *materiality tier* (HIGH / MEDIUM / LOW) at G0, after SR 26-2's exposure-$\times$-purpose logic; the tier sets *who* signs each gate and *how high the pass-bar sits*, but never changes the analytics—every metric is measured identically across datasets. Lowering a tier is itself a recorded G0 decision.

C. Anatomy of a gate

Each gate produces one Gate Decision Record with five parts: (i) the machine recommendation and its numeric basis; (ii) the full evidence pack (the actual numbers, traceable to a run, not a summary); (iii) a closed set of 2–4 options, each carrying its own data, rationale, benefits, risks, and an explicit *residual-risk acceptance*; (iv) a human sign-off (role, identity, exactly one option, free-text rationale, timestamp); and (v) an append-only log entry. The *anti-strawman rule* forbids padding: if the machine cannot construct a genuine case for an alternative, it presents fewer options and flags "single viable option; escalate," routing the decision upward rather than manufacturing fake choices.

Tamper-evidence is a hash chain over the decision records. With a genesis token $h_0 = \mathsf{GEN}\text{-}\langle\text{TAG}\rangle\mathsf{0}$, each gate's entry hash is

$$h_i = \text{SHA-256}\big(h_{i-1} \,\|\, c_i\big)[0{:}8], \tag{4}$$

where c_i is the gate's decision content and $\|$ is concatenation. Any later edit to a sealed decision breaks every subsequent link, so a reversal must be a *new* recorded decision, never a silent edit—the audit trail is append-only.

D. Roles, segregation of duties, and the committee

AEGIS defines seven roles; the hard rule (P2) is that Develop, Validate, and Approve are never the same person for one model, mirroring SR 26-2's conflict-of-interest provisions. Validators sign G1–G4 and can *return* a gate; **G4 requires dual sign-off** (Risk Officer *and* Validator), because calibration and fairness are where model error becomes capital error and conduct risk. The Model Approval Committee—chaired by the CRO or an independent delegate, with a quorum that is void without an independent validator present—rules GF. Any member who developed the model recuses and is minuted; the developer presents but does not vote. Every MAC decision is itself a hash-chained event.

E. The Go/No-Go engine

At GF a deterministic engine synthesises six inputs—dataset adequacy, the signed gate decisions, residual risk vs appetite, the regulatory checklist, audit expectations, and the new-model acceptance bar—into a recommendation the committee rules on: *all gates signed and every regulatory criterion satisfied and residual risk within appetite* $\Rightarrow$ **GO**; *remediable partials with conditions* $\Rightarrow$ **GO-WITH-CONDITIONS** (the engine lists the conditions); *any gap on a must-have criterion, or an unsigned/returned gate, or residual risk outside appetite* $\Rightarrow$ **NO-GO** (the engine names the blocking item and the gate to return to). The engine never upgrades a NO-GO; the committee may downgrade on judgement and must minute why, but cannot clear a hard gap except by resolving it as a new, re-synthesised gate decision.

V. Experimental Setup

Datasets. The corpus is eleven binary-outcome credit datasets (Table II): widely used public benchmarks—including the UCI German and Taiwan credit datasets, the Home Equity (HMEQ) and Give-Me-Some-Credit sets, the FICO HELOC panel, a LendingClub extract, and the PAKDD-2009 Brazilian retail-credit set—alongside additional reference portfolios. They span $N \in [1{,}000,\ 149{,}391]$, prevalence $[6\%, 52\%]$, and candidate pools of 9–87 features. Per Stage 0, protected and proxy attributes are retained with disclosure so results remain comparable with the published literature on these datasets; no model here is intended for live use. Full provenance and target definitions are recorded per dataset.

Fixed configuration (identical for every dataset, by P8). Primary learner `woe_logistic` (XGBoost as a robustness learner); value function cross-validated AUC; correlated columns grouped at $|\rho| = 0.7$; ≥ 10 shadow players with a 95th-percentile bar and confirmation frequency ≥ 0.80; exact Shapley when players (including shadows) ≤ 15, sampled otherwise; selection rule (3); isotonic calibration; PD floor 0.03% and cap 99.9%. The random seed is 42 throughout. No gate re-opens any of these.

Evaluation protocol. Splits are stratified $80/20$ taken *before* any resampling, with the test set held at its natural base rate; the parsimony curve is scored on out-of-fold predictions from stratified 5-fold cross-validation so DeLong's paired test is valid. Discrimination is reported as Gini (primary) with AUC, KS, partial AUC, Brier and ECE; paired model comparisons use DeLong [9]; calibration uses Hosmer–Lemeshow [12] and a per-grade realised-vs-expected backtest; fairness uses the adverse-impact ratio against the four-fifths rule; data sufficiency uses EPV [1] and the Riley minimum-N [2].

Governance and reproducibility. Each dataset is run end-to-end through gates G0–GF with a human sign-off recorded at every gate and appended to the hash chain (4); the materiality tier is set at G0. The pipeline is deterministic: an independent re-run of the German dataset reproduced its committed verdict (Gini 0.53, GO-WITH-CONDITIONS) and hash chain exactly. Validation is i.i.d. cross-validation only—no dataset has a reliable time field, so no claim of out-of-time or production readiness is made—and all accept/decline thresholds and pricing are treated as exogenous institution inputs, never derived here.

TABLE II
THE GOVERNED CORPUS. POOL = FEATURES ENTERING SHARP AFTER STAGE-0; SEL = SELECTED; GINI/AUC ARE OUT-OF-FOLD AT SELECTION. VERDICT IS THE SEALED GF DECISION.

Dataset	N	base	pool→sel	Gini	AUC	Verdict
HMEQ	5,960	.200	12→11	0.81	0.91	GO-cond
CREDITRISK	32,581	.218	9→6	0.76	0.88	GO-cond
GMSC	149,391	.067	9→8	0.70	0.85	GO-cond
HELOC	10,459	.522	23→17	0.59	0.79	GO-cond
TAIWAN	30,000	.221	20→14	0.54	0.77	GO-cond
GERMAN	1,000	.300	17→5	0.53	0.77	GO-cond
MSME	1,707	.060	87→15	0.43	0.72	**NO-GO**
LC	39,717	.142	21→9	0.36	0.68	GO-cond
LOAN	9,578	.160	13→7	0.34	0.67	GO-cond
TH02	1,225	.264	14→6	0.27	0.64	GO-cond
PAK	50,000	.261	39→6	0.15	0.57	**NO-GO**

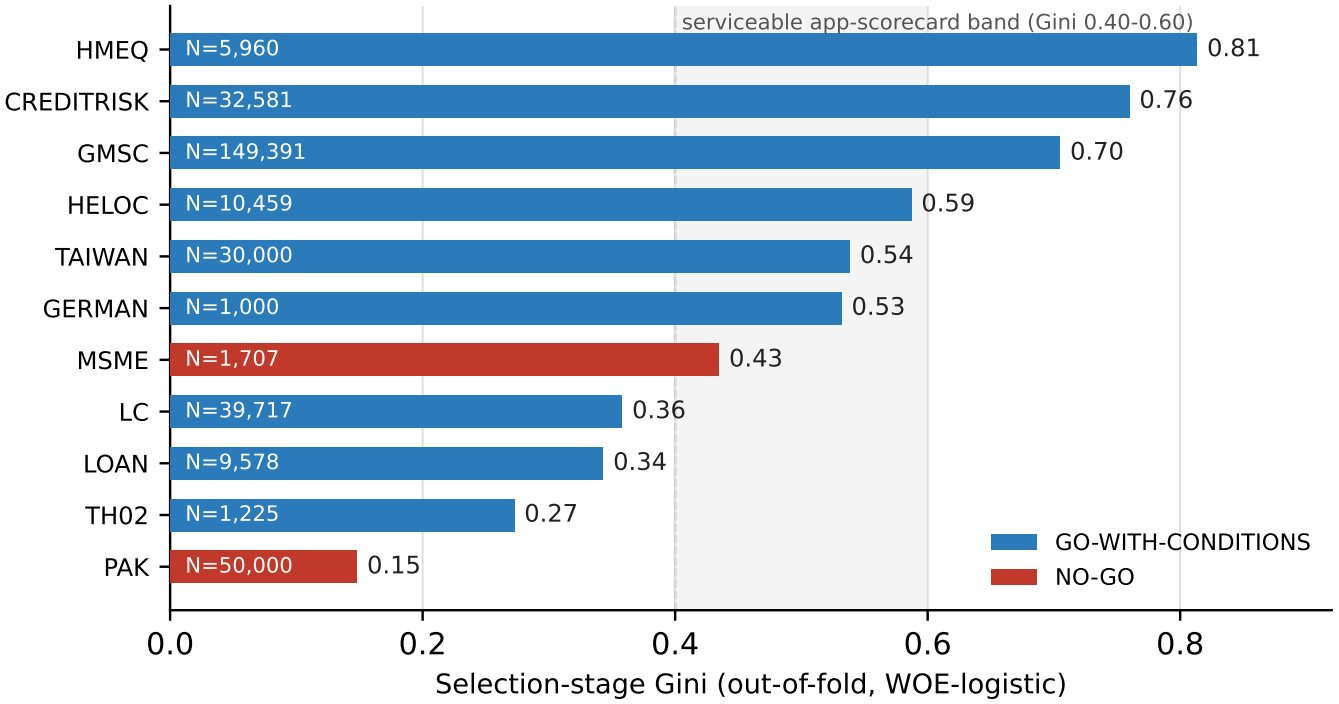


Fig. 1. Selection-stage Gini across the governed corpus, coloured by sealed GF verdict. Discrimination spans an order of magnitude; the two NO-GO runs are not simply the lowest-Gini—MSME is rejected on sufficiency, not discrimination.

VI. RESULTS

We run eleven public and reference credit datasets through the full governed exercise (G0–GF) under *one fixed configuration* (seed 42; selection = smallest k with DeLong $p \geq 0.05$ and $\leq 5\%$ relative-Gini loss; sampled Shapley when players > 15). The corpus spans N from 1,000 to 149,391, prevalence from 6% to 52%, and candidate pools from 9 to 87 features. Table II and Fig. 1 summarise it; nine runs seal GO-WITH-CONDITIONS and two (MSME, PAK) seal NO-GO. Every figure traces to a recorded run.

A. Parsimony varies with sample size

The single fixed stopping rule (3) produces strikingly different parsimony depending on N, because DeLong's power to detect the loss from dropping a weak feature grows with sample size. At **small** N the rule genuinely parsimonises: on German (N=1,000) it selects 5 of 17 features, with both conditions clearing together (DeLong p=0.061 $\geq$ 0.05 and relative-Gini loss 4.10% $\leq$ 5%). At **large** N the same rule returns a near-full set: on LC

TABLE III
RILEY MINIMUM-N RETROSPECTIVE (REPORTED DIAGNOSTIC). $N_{\text{REQ}} = \max(B1, B3, B4)$.

Dataset	N	N_{req}	EPV	Meets?
HMEQ	5,960	278	108	yes
CREDITRISK	32,581	263	1185	yes
GMSC	149,391	570	1251	yes
HELOC	10,459	522	321	yes
TAIWAN	30,000	737	474	yes
GERMAN	1,000	323	60	yes
MSME	1,707	3,654	6.8	**SHORT**
LC	39,717	1,581	625	yes
LOAN	9,578	1,220	219	yes
TH02	1,225	1,170	54	yes (+55)
PAK	50,000	4,041	2173	yes

(N≈40,000) it keeps 8 of 9 players (DeLong p=1.0, loss 0.0%), and on HMEQ (N≈6,000) DeLong rejects *every* proper subset (k=9 vs full, p=6.1×10^{-4}) even though a 5%-loss criterion alone would admit a six-feature model—so the *and*-rule returns all eleven. This is the methodology declining to overstate parsimony when the data can resolve the difference. We call the pattern SIG-N: the same rule, opposite behaviour at the two ends of the sample-size range, both honest. A corollary visible in Table II is that *signal, not volume, sets the ceiling*—German reaches Gini 0.53 on 1,000 rows while LC reaches 0.36 on 40,000.

B. Fairness: excluding a protected attribute is necessary but not sufficient

On two independent datasets, removing the protected attribute did not remove disparate impact, because correlated features reconstruct it. On **LC**, geography (`addr_state`) was dropped at a cost of only 0.007 Gini, yet a residual regional approve-rate disparity persisted (worst adverse-impact ratio 0.62, below the four-fifths bar). On **German**, `age` was excluded at G0, yet the age-band approve-rate AIR was 0.57 (under-25 default rate 48.7% vs 13.3% for 60+); because the model is accurate *within* each band, this is disparate *impact driven by true risk*, not proxy bias. Both runs therefore record the same governed finding—proxy removal is necessary but not sufficient, and a fairness-constrained replication or reject-inference study would be required before any live use—surfaced as a binding GF condition rather than discovered post-deployment.

C. Calibration

With class reweighting off and isotonic calibration always on, calibrated probabilities are tight across the corpus: expected calibration error is 0.020 on HMEQ, 0.012 on LC, and 0.040 on German, and Hosmer–Lemeshow is *not* rejected even at the largest N (GMSC, LC)—the feared large-N hypersensitivity did not materialise for calibration, only for parsimony. Because the learner is WOE-logistic, each PD carries an exact additive reason-code decomposition.

D. Sample-size adequacy (EPV and Riley)

Table III reports the two sufficiency diagnostics. Ten of eleven runs meet the Riley minimum; the lone shortfall is **MSME** (requires 3,654, has 1,707), which is *also* the only run failing EPV ($6.8 < 10$) and the only HIGH-tier NO-GO—the two diagnostics are concordant, and the single failure is the case already rejected. Riley adds what EPV cannot see: **PAK** has an enormous EPV (2,173) yet needs 4,041 rows, because its weak signal (AUC $0.57 \Rightarrow R^2_{cs}$=0.013) makes the shrinkage criterion demand volume; and **TH02** passes by only +55 rows, a thinness EPV (54) does not convey.

VII. DISCUSSION

[Stub: governance vs analytics; exogenous thresholds/pricing; limitations.]

VIII. Conclusion

[Stub.]

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